Comparative Spectral Analysis of Naturalistic Speech and Motor Tasks

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Introduction to the AJILE12

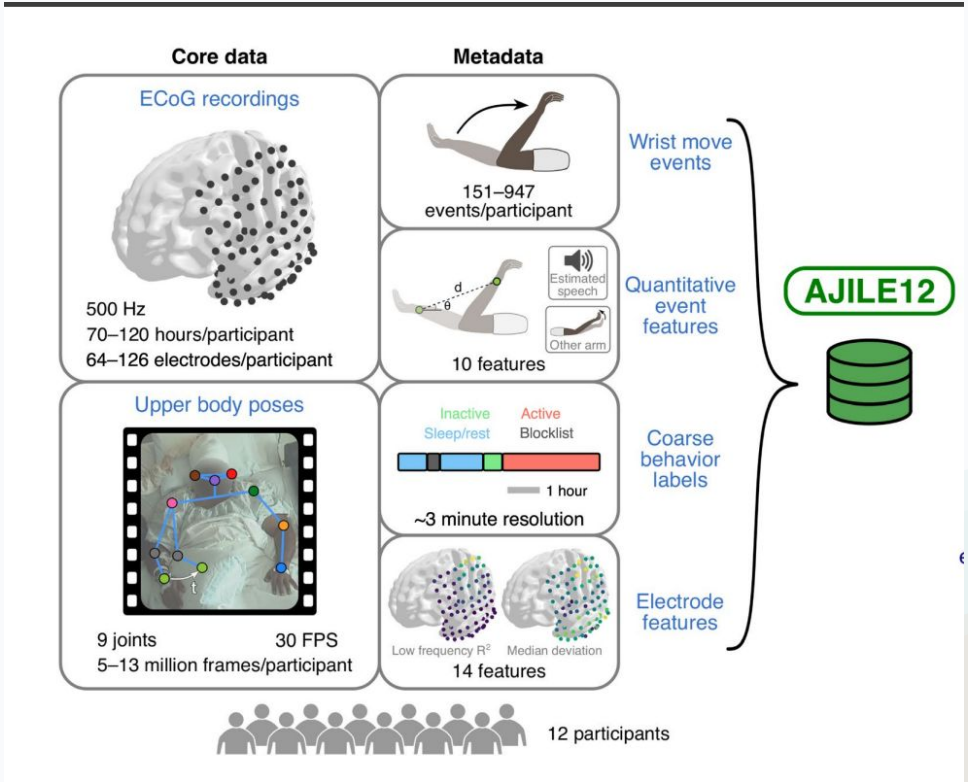
Annotated Joints in Long-Term Electrocorticography Data

Participants

12 human epilepsy participants providing high-fidelity long-term data.

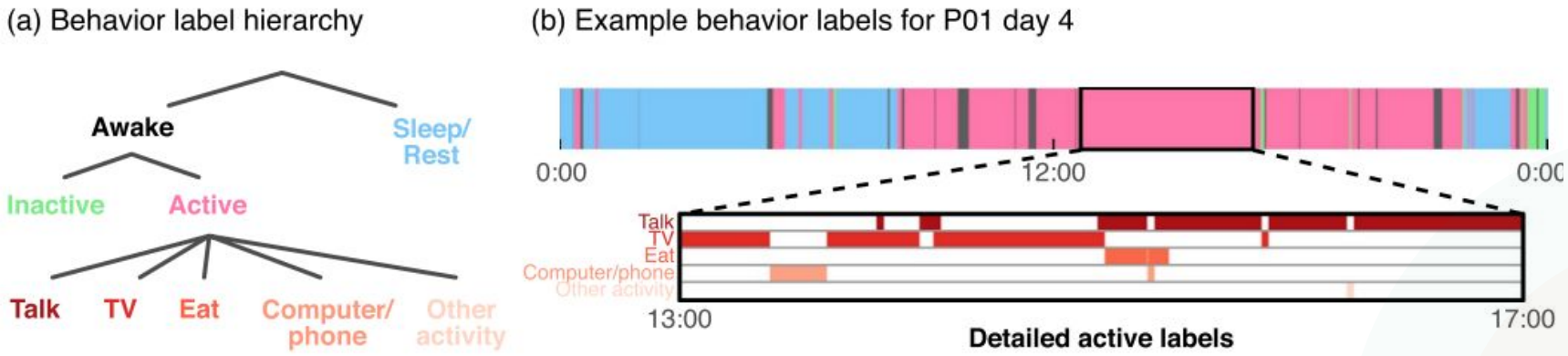
Neural Data

1,280 total hours of synchronized ECoG and pose data for behavioral analysis.



Behavioral Data

Wrist pose trajectories with automated behavior labels: **Talk**, **TV**, **Movement** etc



Aim

Objective

To compare neural power spectral density (PSD) across four distinct behavioral conditions and determine how the addition of speech modifies the neural signatures of a reach event.

Behavioral Conditions

Watch TV (Control)

Passive engagement with minimal motor/speech output.

Speech Only

Neural activity during active vocalization.

Speech + Wrist

Simultaneous execution of motor and language tasks.

Wrist

Execution of motor tasks.

Hypothesis

General Prediction

Since speech centers are typically localized in the **left hemisphere** for most individuals, we predict a strong interaction between speech and motor tasks in **right-handed participants**.

Condition-Specific Prediction

The **"Speech + Wrist"** condition will show higher spectral power or unique interference patterns in the left hemisphere compared to **"Speech Only"** or **"Watch TV."**

Methodology: Electrode Selection Strategy

Challenge: Highly variable electrode placement; standard labels often missing or labeled "unknown".

Solution: Custom MNI-space Distance-Based Mapping Algorithm

Detects primary implanted hemisphere via majority vote on electrode X-coordinates.

MNI Target Coordinates

Motor (Hand/Wrist):

L: [-40, -22, 58] | R: [+40, -22, 58]

Speech :

L: [-54.0, -6.0, 28.0] | R: [54.0, -6.0, 28.0]

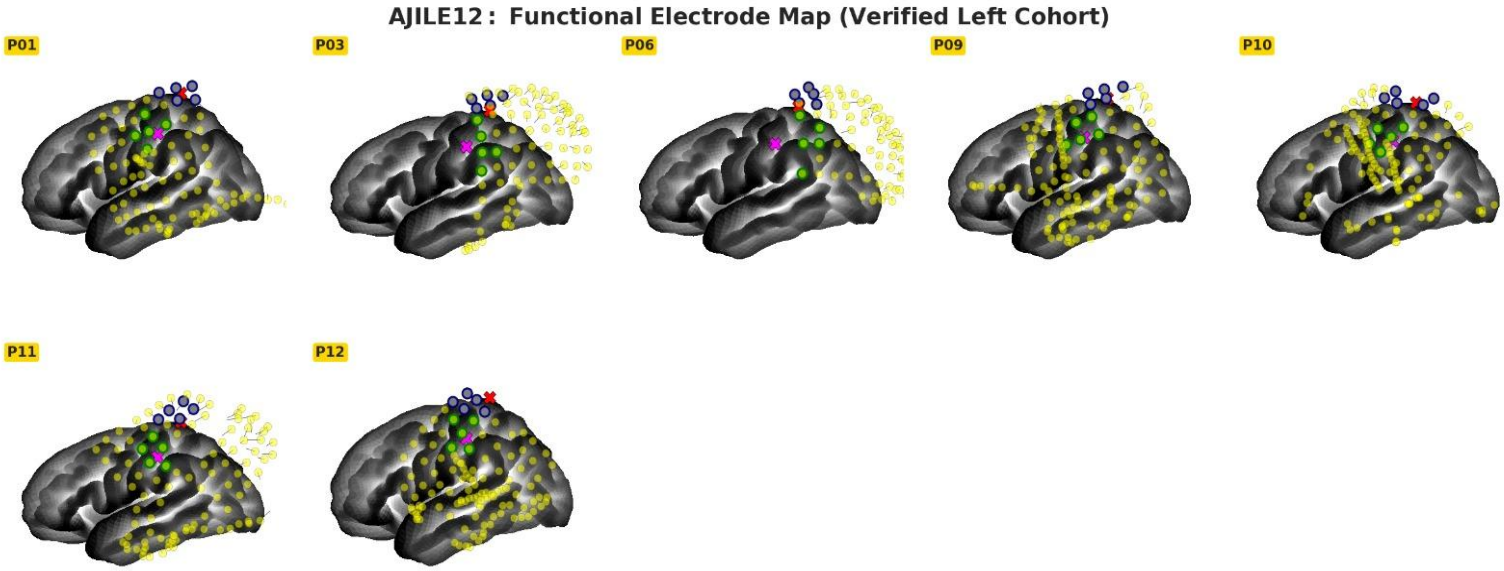
Selection Criteria

Top 3 closest good electrodes per target.

- **Motor:** Surface grids
- **Speech:** Ventral Sensorimotor Cortex

Yousry, T. A., Schmid, U. D., Alkadhi, H., Schmidt, D., Peraud, A., Buettner, A., & Winkler, P. (1997). Localization of the motor hand area to a knob on the precentral gyrus: A new landmark. *Brain*, 120(1), 141-157

Bouchard, K. E., Mesgarani, N., Johnson, K., & Chang, E. F. (2013). Functional organization of human sensorimotor cortex for speech articulation. *Nature*, 495(7441), 327-332.


AJILE12 : Functional Electrode Map (Verified Left Cohort)
P01
P03
P06
P09
P10
P11
P12

Methodology – Data Extraction Pipeline

Logic

Clean behavioral conditions are isolated from ECoG data using a fixed window of 2.0 seconds with a 0.5-second step size.

Only snippets with exact sample length (at 500 Hz sampling rate) are retained.

Condition Partitioning

Baseline (TV/Idle)

Extracted from coarse epoch metadata.

Isolated Speech

Talking periods excluding any motor movement labels.

Motor & Interaction

- High-Resolution Motor: Reach intervals split.
- Move Only.
- Both (Speech + Move): 10% speech threshold.

Spectral Analysis Pipeline: Welch's Method

1. Segmentation

Signal x(n) partitioned into K overlapping segments of length M. Windowed by w(n) (e.g., Hann) to mitigate spectral leakage.

$$X_i(f) = \sum_{n=0}^{M-1} w(n) x_i(n) e^{-j2\pi fn}$$

2. Periodogram

Power calculated via squared magnitude. Normalization factor U maintains energy conservation.

$$P_i(f) = \frac{1}{M \cdot U} |X_i(f)|^2$$

$$\text{where} U = \sum_{n=0}^{M-1} w^2(n)$$

3. Averaging

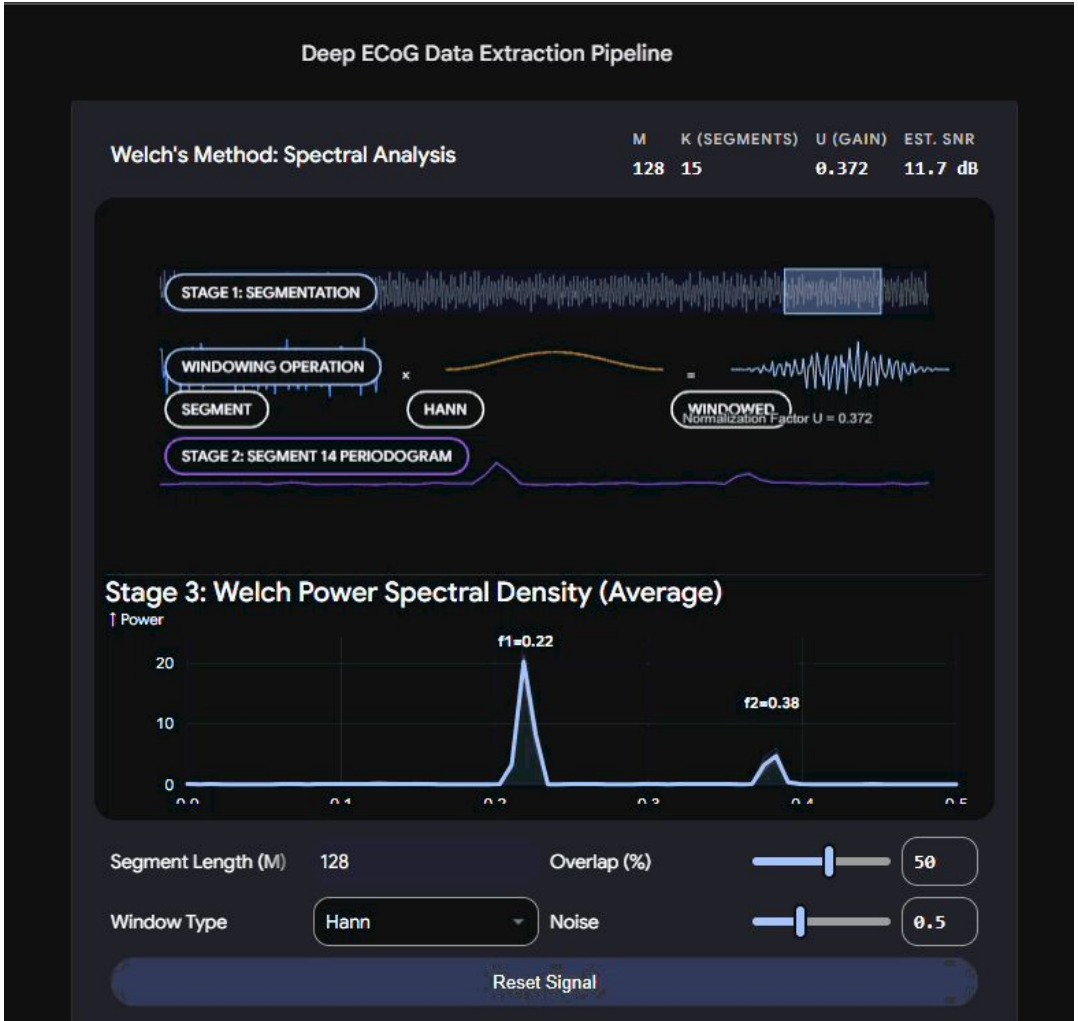
Final PSD is the ensemble average, converting inconsistent data into a smooth, reliable estimator.

$$\hat{P}_{Welch}(f) = \frac{1}{K} \sum_{i=1}^{K} P_i(f)$$

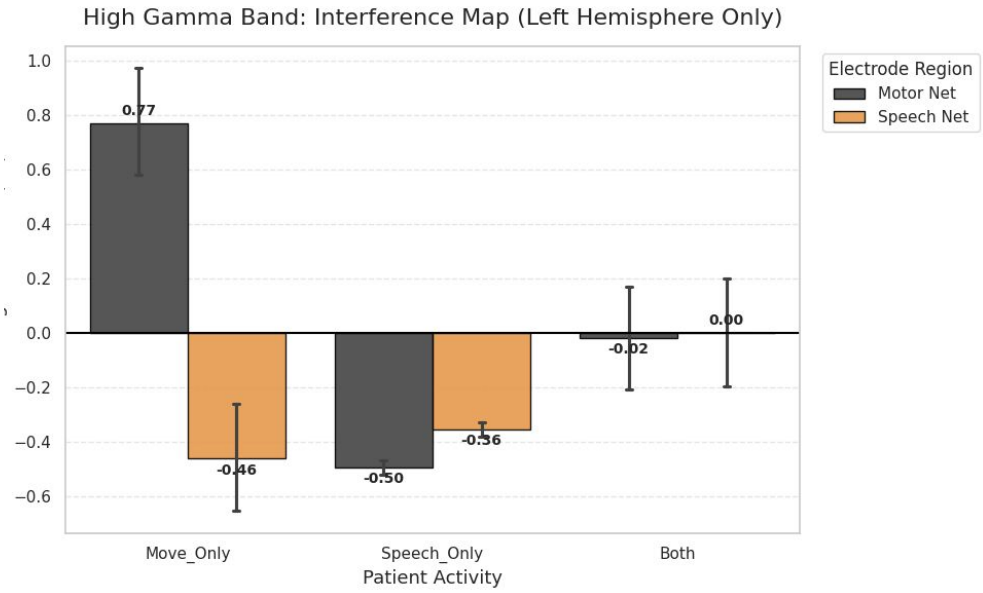
Why Welch's Method was Chosen

Variance Reduction: Reduces noise variance by factor K, revealing subtle **+0.77 dB High Gamma activation** obscured in standard FFTs.

Robustness to Non-Stationarity: Short, overlapping windows better capture transient brain activity during tasks like reaching or speaking.


Deep ECoG Data Extraction Pipeline
Welch's Method: Spectral Analysis
M
128
K (SEGMENTS)
15
U (GAIN)
0.372
EST. SNR
11.7 dB
STAGE 1: SEGMENTATION
WINDOWING OPERATION
SEGMENT
HANN
WINDOWED
Normalization Factor U = 0.372
STAGE 2: SEGMENT 14 PERIODOGRAM
Stage 3: Welch Power Spectral Density (Average)
↑ Power
f1=0.22
f2=0.38
20
10
0
Segment Length (M)
128
Overlap (%)
50
Window Type
Hann
Noise
0.5
Reset Signal

Results and Analysis



Key Observations

Isolated Motor Activation

During Move_Only, the Motor Network shows a robust High Gamma power increase of **+0.77 dB**.

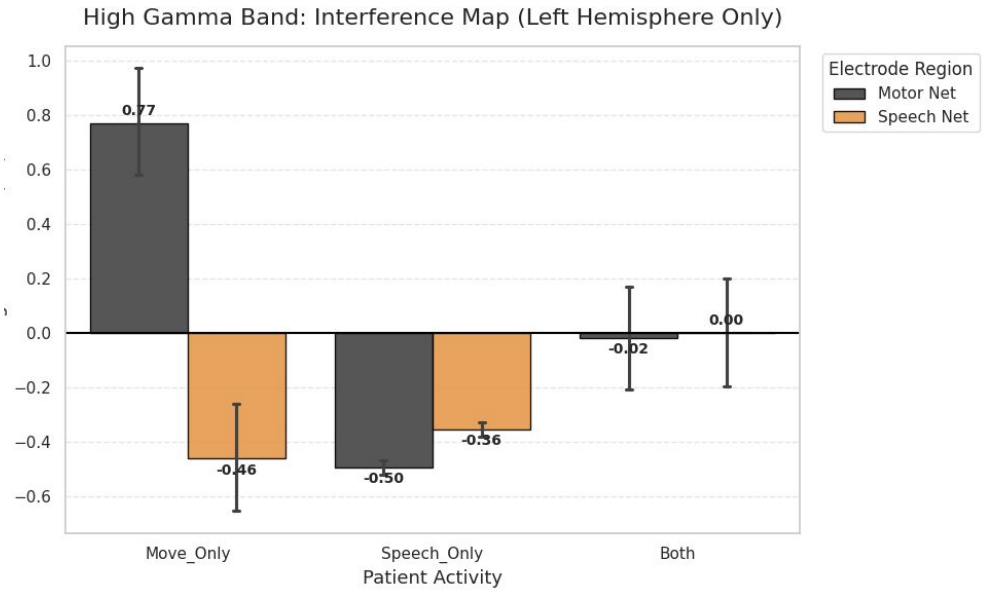
The Baseline Shift

Speech_Only results in a **-0.36 dB** power drop in the Speech Network relative to baseline.

Signal Neutralization & Variability

In Both condition, Motor power collapses to **-0.02 dB**. Error bars indicate higher variance.

Results and Analysis



Conclusions

Speech as a "Signal Killer"

High Gamma motor signals are not additive; instead, speech acts as a primary interferent that obliterates movement signatures.

The "TV Baseline" Paradox

Negative values during speech confirm that passive dialogue processing (watching TV) is more electrophysiologically demanding than active vocalization.

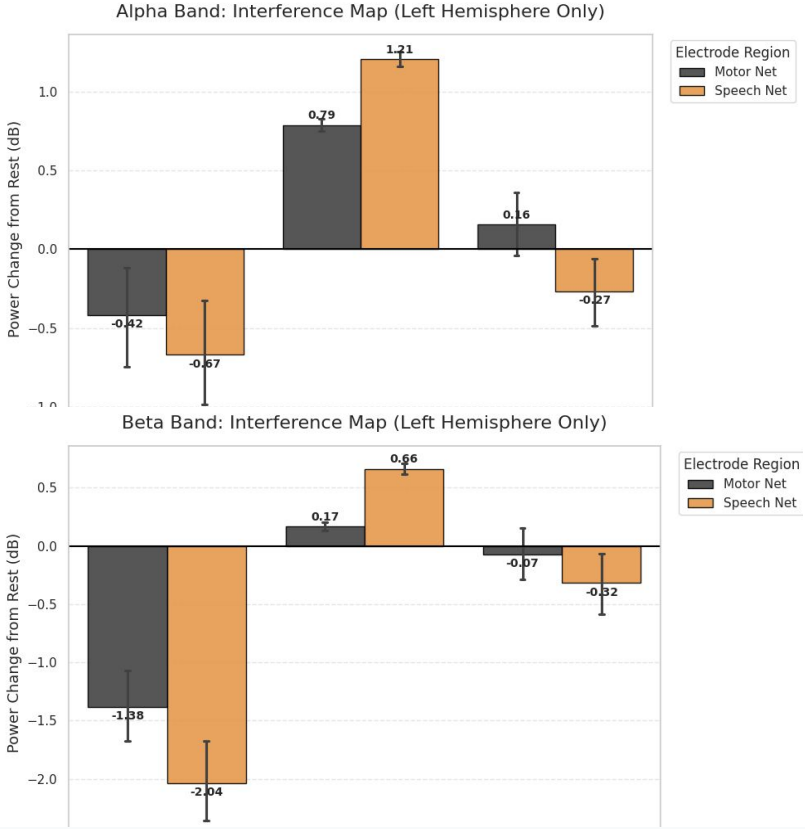
Decoding Blindness

Standard BCI decoders calibrated for isolated movement become "blind" to user intent during conversation.

Need for Robust Architectures

Decoders must incorporate spatial filtering or multi-band integration to isolate intent from cross-talk.

Results and Analysis



Observations

Task-Specific Desynchronization (`Move_Only`)

A significant power attenuation in both bands (~ -1.0 dB). Represents "release" of cortical inhibition.

Functional Synchronization (`Speech_Only`)

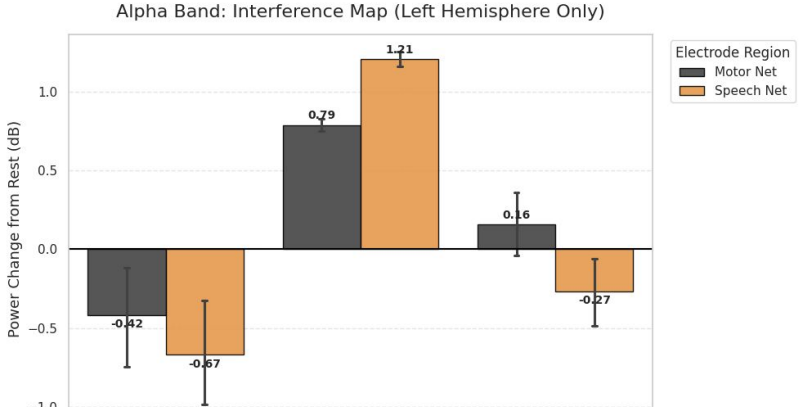
A marked power increase (~ +0.9 dB) within motor network, indicating active suppression of movement.

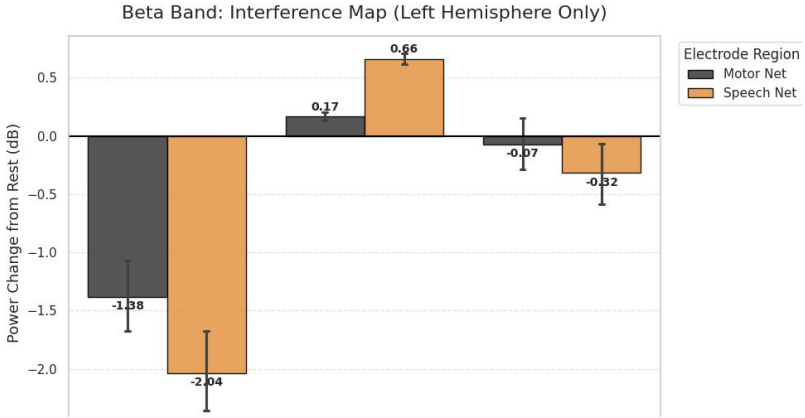
Signal Neutralization (`Both`)

Concurrent tasks result in near-zero shift (~ 0 dB). Speech inhibition cancels excitatory motor command.

Engel, A. K., & Fries, P. (2010). "Beta-band oscillations—signalling the status quo?" Current Opinion in Neurobiology.
Jensen, O., & Mazaheri, A. (2010). "Shaping functional architecture by oscillatory alpha activity: gating by inhibition." Frontiers in Human Neuroscience.
Pfurtscheller, G., & Lopes da Silva, F. H. (1999). "Event-related EEG/MEG synchronization and desynchronization: basic principles." Clinical Neurophysiology.

Results and Analysis





Conclusions

Neural "Braking" System

Alpha and Beta rhythms function as inhibitory brakes; speech acts as a "locking" mechanism that physically prevents the motor cortex from transitioning to an active state.

Dual-Task Interference

Simultaneous speech and movement create a state of competition where the brain's inhibitory gating overrides the motor intent.

Decoding Bottleneck

BCI systems relying on low-frequency power drops will fail during natural conversation, as the speech-induced power spike masks the intentional movement signal.

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Jensen, O., & Mazaheri, A. (2010). "Shaping functional architecture by oscillatory alpha activity: gating by inhibition." Frontiers in Human Neuroscience.
Pfurtscheller, G., & Lopes da Silva, F. H. (1999). "Event-related EEG/MEG synchronization and desynchronization: basic principles." Clinical Neurophysiology.

Conclusion

The Neuro-Electrical Impact of Speech on Motor Intent

The introduction of speech initiates a physiological override that fundamentally alters the electrical signatures of the motor cortex through three primary mechanisms:

1. Signal Masking

(High Gamma Erasure)

Intentional movement spikes High Gamma power (**+0.77 dB**). Speech obliterates this signal, flatlining it to **-0.02 dB**. The linguistic demand "drowns out" the motor command.

2. Inhibitory Conflict

(Alpha/Beta Jamming)

Vocal control triggers a **Postural Lock** spike that opposes the motor "release" drop..

3. Resource Competition

(Cortical Recruitment)

The brain prioritizes vocal networks over limb movement. This lead to **attenuated recruitment**, resulting in shallower, variable signals that are difficult to decode.

Summary: "Total Cortical Washout" renders traditional BCI algorithms ineffective during conversation.

Challenges – Computational Bottlenecks

Memory & Data Fragmentation

Memory overload and data fragmentation from loading full 24-hour 500 Hz ECoG recordings caused RAM crashes and biased PSD results.

Latency & Alignment Issues

High latency from DANDI streaming, sampling mismatches, and snippet alignment issues slowed debugging and analysis.

Workarounds

Shape Enforcer + Resume Logic

Ensured exact 2.0s (1000-sample) windows and skipped already-processed participants to prevent redundant computation and enable crash recovery.

Aggressive RAM & Storage Management

Used gc.collect(), immediate float32 conversion, explicit del, and compressed .npz files to minimize memory usage and enable efficient processing.

Future scope and application

Enhanced BCI Systems

Insights into speech-motor interference will enable more robust Brain-Computer Interfaces that support reliable multi-tasking (speech + movement) in real-world environments.

Clinical Diagnostics

Long-term naturalistic ECoG data can improve understanding of neurobehavioral variability and recovery patterns in epilepsy patients.

Adaptive & Context-Aware BCIs

Develop systems that automatically detect behavioral states (e.g., talking vs. moving) and dynamically switch decoding strategies for seamless, generalizable performance.

Restorative Neurotechnology

Translate findings into speech neuroprostheses and rehabilitation tools for patients with communication or motor impairments (e.g., ALS, stroke, or locked-in syndrome).

Thank You!

Pseudocode: Data Extraction Logic

```
Initialize Parameters: Motor_Target, Speech_Target, Window_Size, Speech_Threshold (0.1)

FOR each Patient in Cohort:
Identify Electrode Indices (Find 5 nearest to Motor and Speech targets)

FOR each Asset (NWB File) in Patient Data:
Load Interval Tables: 'epochs' (Macro) and 'reaches' (Micro)

# --- PHASE 1: MACRO FILTERING (TV & PURE SPEECH) ---
FOR each row in 'epochs':
IF label contains "EATING": Add to Blacklist_Intervals
IF label contains "TV" or "IDLE": Condition = "TV"
ELSE IF label contains "TALK" and NOT "MOVE": Condition = "Speech_Only"; Add to Speech_Intervals
IF Condition is Valid: Extract 2s snippets every 0.5s via SLIDING WINDOW

# --- PHASE 2: MICRO FILTERING (MOVE & CROSS-TALK) ---
FOR each row in 'reaches':
IF reach overlaps with Blacklist_Intervals: SKIP
IF reach overlaps with Speech_Intervals OR speech_ratio > Threshold: Condition = "Both"
ELSE: Condition = "Move_Only"
IF reach_duration < 2s: PAD to 2s; ELSE: SLIDING WINDOW
Extract Raw ECoG -> Save to Condition_Buffer
```

https://github.com/Saimapath/BB-427-endsem.git